

Function, Domain & Range

1. a) Define function with example. Describe the various types of function. (Au-25)
 b) Test whether the following functions are even or odd:

i. $f(x) = 1 - \frac{1}{x^2}$

ii. $f(x) = x + \frac{1}{x}$

iii. $f(x) = \ln(x + \sqrt{1+x^2})$

2. a) Define domain & range of a function with example. Imp.

- b) Find the domain & range of the following function:

i. $f(x) = x+3$

ii. $f(x) = x^2+x+2$

iii. $f(x) = \sqrt{x+3}$

iv. $f(x) = \sqrt{16-x^2}$

v. $f(x) = \sqrt{x^2-9}$ (Au-25)

vi. $f(x) = \sqrt{\frac{x+2}{x-2}}$ (Sp-26)

vii. $f(x) = \sqrt{\frac{1-x}{x}}$ (Sp-25)

viii. $f(x) = \frac{x+1}{x-1}$

ix. $f(x) = \frac{x-3}{2x+1}$

x. $f(x) = \frac{x}{|x|}$

xi. $f(x) = |x| - x$

xii. $f(x) = |x+1| + |x-1|$

xiii. $f(x) = |x| + |x-1|$

xiv. $f(x) = |x+2| + |x-2|$ imp

xv. $f(x) = |x+3| + |x-3|$ (Sp-24)

xvi. $f(x) = |x+1| + |x| + |x-1|$

xvii. $f(x) = \begin{cases} 3, & -3 \leq x < -1 \\ -6x-3, & -1 \leq x \leq 0 \\ 3x-3, & 0 < x \leq 1 \end{cases}$

→ Piecewise function

3. Draw the graph of the following function:

i. $f(x) = \sqrt{x}$

ii. $f(x) = \frac{1}{x}$

iii. $f(x) = -x^2 + 1$

iv. $f(x) = 3 - 2x - x^2$

v. $f(x) = \begin{cases} x^2 + 1, & x < 0 \\ x, & 0 \leq x \leq 1 \\ \frac{1}{x}, & x > 1 \end{cases}$

vi. $f(x) = \sin x, \cos x, \tan x$

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vii. $f(x) = a^x, a > 1$
 viii. $f(x) = |x| + |x-1|$
 ix. $f(x) = \begin{cases} 3+2x, & -\frac{3}{2} < x < 0 \\ 3-2x, & 0 \leq x < \frac{3}{2} \\ -3-2x, & x \geq \frac{3}{2} \end{cases}$

Limit

1. Define limit of a function.
 2. a) If $f(x) = \begin{cases} 1+2x, & -\frac{1}{2} \leq x < 0 \\ 1-2x, & 0 \leq x < \frac{1}{2} \\ 2x-1, & x \geq \frac{1}{2} \end{cases}$
 Does limit $\lim_{x \rightarrow 0} f(x)$ and $\lim_{x \rightarrow \frac{1}{2}} f(x)$ exists?

b) If $f(x) = \begin{cases} x^2, & x < 1 \\ 3, & x = 1 \\ x^2 + 1, & x > 1 \end{cases}$

Does limit $\lim_{x \rightarrow 1} f(x)$ exists?

c) If $f(x) = \begin{cases} x, & x > 0 \\ 0, & x = 0 \\ -x, & x < 0 \end{cases}$

Does limit $\lim_{x \rightarrow 0} f(x)$ exists?

d) Let, $f(x) = \frac{3x+|x|}{7x-5|x|}$ does limit $\lim_{x \rightarrow 0} f(x)$ exists?

Continuity

1. Define continuity of a function.
 2. Test the continuity of the following function:

$f(x) = \begin{cases} 5, & 0 < x < 1 \\ 10, & 1 \leq x \leq 2 \\ 15, & 2 < x \leq 3 \end{cases}$ at $x = 2$

3. Test the continuity of the following function:

$f(x) = \begin{cases} \frac{1-\cos x}{x^2}; & x \neq 0 \\ 1; & x = 0 \end{cases}$ at $x = 0$ (Au-25, Sp-26)

4. Test the continuity of the following function at $x = 0$ and $x = \frac{3}{2}$

$f(x) = \begin{cases} 3+2x, & -\frac{3}{2} \leq x < 0 \\ 3-2x, & 0 \leq x < \frac{3}{2} \\ -3-2x, & x \geq \frac{3}{2} \end{cases}$ (Sp-24)

5. Test the continuity at $x = 0$ and $x = 1$

$$f(x) = \begin{cases} x^2 + 1, & x < 0 \\ x, & 0 \leq x \leq 1 \\ \frac{1}{x}, & x > 1 \end{cases}$$

6. Test the continuity at $x = 0, 1, 2$.

$$f(x) = \begin{cases} -x^2, & x \leq 0 \\ 5x - 4, & 0 < x \leq 1 \\ 4x^2 - 3x, & 1 < x < 2 \\ 3x + 4, & x \geq 2 \end{cases}$$

7. Test the continuity at $x = 0$

$$f(x) = \begin{cases} \frac{|x|}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$$

8. Test the continuity at $x = 0$

$$f(x) = \begin{cases} -\cos x, & x \neq 0 \\ \cos x, & x = 0 \end{cases}$$

9. Test the continuity at $x = 0$

$$f(x) = \begin{cases} \frac{\log_e(1+x)}{\tan^{-1} x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$$

10. Test the continuity at $x = 1, 2$

$$f(x) = \begin{cases} \log x, & 0 < x \leq 1 \\ 0, & 1 < x < 2 \\ 1 + x^2, & x \geq 2 \end{cases}$$

11. Test the continuity of the function, $f(x) = |x| + |x + 1|$ at $x = 1$

Differentiability

1. Define differentiability of a function. *impl*

2. Test whether the function $f(x) = x^2 + 5$ is differentiable at the point $x = 1$.

3. A function $f(x)$ is defined as;

$$f(x) = \begin{cases} 5x + 4, & 0 < x \leq 1 \\ 4x^2 - 3x, & 1 < x < 2 \end{cases}$$

i) Test the continuity at $x = 1$

ii) Test whether the above function is differentiable at the point $x = 1$.

4. Test the differentiability of the following function at $x = 0$

$$f(x) = \begin{cases} 3 + 2x, & -\frac{3}{2} < x < 0 \\ 3 - 2x, & 0 \leq x < \frac{3}{2} \\ -3 - 2x, & x \geq \frac{3}{2} \end{cases} \quad (\text{Sp-24, Sp-26})$$

5. Test the differentiability of the function at $x = \frac{\pi}{2}$

$$f(x) = \begin{cases} 1, & x < 0 \\ 1 + \sin x, & 0 \leq x < \frac{\pi}{2} \\ 2 + (x - \frac{\pi}{2})^2, & x \geq \frac{\pi}{2} \end{cases} \quad (\text{Sp-24})$$

6. Test the differentiability of the function at $x = 2$

$$f(x) = \begin{cases} 3x + 2; & x > 2 \\ 12 - x^2; & x \leq 2 \end{cases} \quad (\text{Sp-25})$$

7. Test the differentiability of the function at $x = 2$

$$f(x) = \begin{cases} x; & 0 < x < 1 \\ 2 - x; & 1 \leq x \leq 2 \\ x - \frac{1}{2}x^2; & x > 2 \end{cases} \quad (\text{Sp-25})$$

Indeterminate Form

1. $\lim_{x \rightarrow 0} \frac{xe^x - \log(1+x)}{x^2}$

2. $\lim_{x \rightarrow 0} \frac{e^x - e^{-x} + 2\sin x - 4x}{x^5}$

3. $\lim_{\theta \rightarrow \frac{\pi}{4}} \frac{\sqrt{2} - \cos \theta - \sin \theta}{(4\theta - \pi)^2}$

4. $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a}$

5. $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$

6. $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$

7. $\lim_{x \rightarrow a} \frac{x^{\frac{7}{2}} - a^{\frac{7}{2}}}{\sqrt{x} - \sqrt{a}}$

8. $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$

9. $\lim_{x \rightarrow 0} x^2 \log x^2$

10. $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}. (\text{Sp-24})$

11. Evaluate the limit using L. Hospital's rule $\lim_{x \rightarrow \infty} 2^x \sin\left(\frac{a}{2^x}\right). (\text{Sp-26})$

12. Evaluate the limit using L. Hospital's rule $\lim_{x \rightarrow 0} \left(\frac{1}{x}\right)^{\tan x}$. (Sp-25)
13. Evaluate the limit using L. Hospital's rule $\lim_{x \rightarrow \infty} (e^x + 1)^{\frac{-2}{x}}$. (Au-25)

Differentiation

- Describe the physical meaning of derivative of a function.
- The radius of a circle increases at a rate of 3 cm/sec. Find the rate of change of the area when i) the radius is 5 cm. ii) the area is 4π cm² (Sp-24)
- By definition (First principal method), find the differential co-efficient of the following function:
 - $f(x) = x^n, a^x, \ln x$ (Au-25), $\log_a x$ (Sp-26)
 - $f(x) = e^x$ (Sp-25), $e^{mx}, e^{\sqrt{x}}$
 - $f(x) = \sin x$ (Sp-24), $\cos x, \tan x, \cot x, \sec x, \csc x$
 - $f(x) = \sin^{-1} x, \cos^{-1} x, \tan^{-1} x, \cot^{-1} x, \sec^{-1} x, \csc^{-1} x$
- Differentiate the following functions or derive the first derivatives from the following functions:
 - $y = \sec \tan^{-1} x$
 - $y = \log \sec(ax + b)^3$ (Sp-24)
 - $y = \ln(x^2 + 2)$
 - $y = 5x^{\frac{3}{2}} + \tan^2(2x)$
 - $y = \log_a x + \log_x a$
 - $y = (\ln \sin x)^2$
 - $y = \frac{x^n}{\log_a x}$
 - $y = \tan^{-1} \frac{a+bx}{b-ax}$
 - $y = \sqrt[3]{x^2 + \sqrt{x}}$
 - $y = 10^{\ln \sin x}$
 - $y = e^{\sin x} - (\sin x^2)^2$
 - $y = x^{x^x}, x^{e^x}, (\sin x)^{\ln x}, x^{\cos^{-1} x}, (\sin^{-1} x)^{\ln x}$ (Sp-26)
 - $y = \sin^2(\ln x^2)$
 - $y = (\sec \tan^{-1} x)(x^5 + x^{\frac{1}{3}})^{\frac{5}{3}}$
 - $y = \sin(2 \tan^{-1} \sqrt{\frac{1-x}{1+x}})$
 - $y = \sin^{-1} \left(\frac{x + \sqrt{1-x^2}}{\sqrt{2}} \right)$
 - $\sqrt{\frac{x}{y}} + \sqrt{\frac{y}{x}} = 1$ (Sp-26)
- If $y = \sqrt{\cos x + \sqrt{\cos x + \sqrt{\cos x + \dots \dots \dots \infty}}}$ then find $\frac{dy}{dx}$.
- If $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \dots \dots \dots \infty}}}$; then show that, $y_1 = \frac{\cos x}{2y-1}$ (Sp-25)

7. If $y = \cos x \cos x \cos x \dots$ then find $\frac{dy}{dx}$ (Au-25)

8. Differentiate $\tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$ with respect to $\sec^{-1} \frac{1}{2x^2-1}$ (Au-25, Sp-26)

9. Differentiate $\tan^{-1} \frac{2x}{1-x^2}$ with respect to $\sin^{-1} \frac{2x}{1+x^2}$ (Sp-24, Sp-25)

10. If $C = 1 + r \cos \theta + \frac{r^2 \cos 2\theta}{2!} + \frac{r^3 \cos 3\theta}{3!} + \dots$
 $S = r \sin \theta + \frac{r^2 \sin 2\theta}{2!} + \frac{r^3 \sin 3\theta}{3!} + \dots$

Prove that, $C \frac{dC}{dr} + S \frac{dS}{dr} = (C^2 + S^2) \cos \theta$

11. The radius of a circle increases at rate of 3 cm/sec. Find the rate of change of the area when i) the radius is 5 cm ii) the area is $4\pi \text{ cm}^2$. (Sp-24)

Successive Differentiation

1. Find the n^{th} derivative of the following functions;

i. $y = \ln x, \frac{1}{x}, a^x, e^{mx}, x^m, \sin x, \cos x, \ln(ax+b), \sin(ax+b), \cos(ax+b)$

2. State and prove Leibnitz's theorem.

3. If $y = x^n \log x$; prove that, $y_{n+1} = \frac{n!}{x}$. (Sp-24)

4. If $y = \tan^{-1} x$ then show that, $(1+x^2)y_{n+1} + 2xy_n + n^2y_{n-1} - ny_{n-1} = 0$; (Sp-25)

5. If $x = \cos \sqrt{y}$ then show that, $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$. (Sp-26)

6. If $y = e^{a \cos^{-1} x}$ or $y = e^{a \sin^{-1} x}$ then show that, $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+a^2)y_n = 0$

7. If $x = \sin(\frac{1}{m} \ln y)$ then show that, $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+m^2)y_n = 0$ (Au-25)

8. If $\ln y = \tan^{-1} x$, then show that, $(1+x^2)y_{n+2} + (2nx + 2x - 1)y_{n+1} + n(n+1)y_n = 0$

9. If $y = a \sin^{-1} x + b \cos^{-1} x$ then show that, $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$

10. If $y = f(x) = \sin^{-1} x$ then show that,

i. $(1-x^2)y_2 - xy_1 = 0$

ii. $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$

iii. $(y_n)_0 = (n-2)^2 \cdot (n-4)^2 \dots \dots \dots 3^2 \cdot 1^2$ for n is odd number.

iv. $(y_n)_0 = 0$ for n is even number.

General Theorem & Expansion

Rolle's Theorem

1. State and prove Rolle's theorem.

2. Verify Rolle's theorem, $f(x) = 2x^3 + x^2 - 4x - 2$

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3. Verify the Rolle's theorem for the function $f(x) = x^2 - 3x + 2$ in the interval $(1, 2)$. (Au-25)
4. Suppose we are asked to determine whether Rolle's theorem can be applied to $f(x) = x^4 - 2x^2$ on the closed interval $[-2, 2]$. And if so, find all values of c in the interval that satisfies the theorem. (Sp-24)
5. Verify the Rolle's theorem for $f(x) = 2x^3 + x^2 - 4x - 2$ over $[-\sqrt{2}, \sqrt{2}]$; (Sp-25)
6. Verify the Rolle's theorem for $f(x) = e^x \sin x$ in the interval $(0, \pi)$. (Sp-26)

Mean Value Theorem

1. State and prove **Mean Value** theorem.
2. Verify the mean value theorem for the function, $f(x) = 3 + 2x - x^2$ in the interval $(0, 1)$; (Sp-25)
3. Verify the mean value theorem for the function, $f(x) = 3x^3 + 7x^2 - 11x - 15 = 0$
4. Verify the mean value theorem for the function, $f(x) = \frac{x^3}{4} + 1$ over the interval $[0, 2]$. (Au-25)
5. Verify the mean value theorem for $f(x) = 2x^3 - 8x + 1$ when $a = 1$ and $b = 3$. (Sp-24)
6. Verify the mean value theorem for the function, $f(x) = x^2 - 5x + 7$ in the interval $1 \leq x \leq 3$.
7. Verify the mean value theorem for the function, $f(x) = x(x-1)(x-2)$ in the interval $(0, 0.5)$. (Sp-26)

Taylor's Series:

1. State Taylor's series.
2. Expand $\frac{1}{x}$ in ascending power of $(x-2)$ by Taylor's series. (Au-25)
3. Expand $\ln x$ in ascending power of $(x-2)$ by Taylor's series. (Sp-26)

Maclaurin's Series

8. State and prove **Maclaurian's** theorem.
9. Obtain the Maclaurian's series generated by the function, $f(x) = \log(1+x)$ (Sp-24, Au-25), e^x , e^{mx} , $\sin x$, $\cos x$ (Sp-25), $\ln x$ etc.

Recommendation Book:

1. A text book on differential calculus - P. K. Bhattacharjee
2. Differential calculus - P. N. Chatterjee